Homework 1

ASEN 5335 Aerospace Environment

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Problem 1: Radiation from Corona

(i)

Given T of the corona region to be $T = 2MK$. We can use the equation,

$$\lambda_{max} T = \alpha$$

Where $\alpha = \text{constant}(2.98 * 10^-3K)$.

Rearrange to solve for wavelength.

$$\lambda_{max} = \frac{T}{\alpha}$$

The corona region's temperature can be substituted and solved.

$$\boxed{\lambda_{max} = 1.449 * 10^-9\,\text{m} = 14.49 * 10^-10\,\text{m}}$$

This would also be in the same range for X-rays, $\approx 10^-10\,\text{m}$.

(ii)

For the region photosphere region at a cooler $T = 5,778\,\text{K}$, the process is the same.

$$\boxed{\lambda_{max} = 5.01 * 10^-7\,\text{m} = 50.15 * 10^-8\,\text{m}}$$

This would be in the same range for UV $\approx 10^-8\,\text{m}$.

Problem 2: Thermal Escape

From Wikipedia we can obtain an equation for most probable thermal velocity of a particle.

$$v_{\text{th}} = \sqrt{\frac{2k_B T}{m}}$$

$$k_B = 1.38 \times 10^{-23}\,\text{J/K}$$

$$m = m_{\text{proton}} = 1.673 \times 10^{-27}\,\text{kg}$$

$$T = 2\,\text{MK} = 2 \times 10^6\,\text{K}$$

$$\boxed{\therefore v_{th} = 181,644\,\text{m/s}}$$

We then calculate the escape velocity using the equation,

$$v_{escape} = \sqrt{\frac{2GM}{r}}$$
$$G = 6.674 * 10^{-11}\,\mathrm{Nm^2/kg^2}$$
$$M = 1.989 * 10^{30}\,\mathrm{kg}$$
$$r = 10 * 10^6 * 10^3\,\mathrm{m}$$

Where G is the unversial gravitational constant, M is the mass of the sun, and r is the radius of the of the outer corona. We then find,

$$\boxed{v_{escape} = 163,000\,\mathrm{m/s}}$$

$$\because v_{th} < v_{escape}$$
$$\boxed{\text{It is possible for the coronal protons to escape the sun's gravity}}$$

Problem 3

(a)

Write your solution here.

(b)

Write your solution here.

Problem 4

Write your solution here.